NCFE Level 2 Functional Skills Qualification in Mathematics (603/5060/X)

Paper number:	SAM
Section A:	Non-calculator Test



Time allowed: 30 minutes

Learner instructions

- Answer **all** questions.
- Read each question carefully.
- Write your answers in the spaces provided.
- Show your working, as marks may be awarded for working.
- State units in your answers, where appropriate.
- Check your work.

Learner information

- Section A contains **Activity 1** only.
- The maximum mark for this section is **15**.
- The marks available for **each** question are shown in brackets.

To be completed by the examiner		Mark
A	Activity 1	/ 15
B	Activity 2	/ 15
	Activity 3	/ 15
	Activity 4	/ 15
TOTAL MARK		/ 60

Resources

You will need a:

- pen, with black or blue ink
- pencil and eraser
- 30 cm ruler
- protractor.

If extra pages are used, please make sure your name and centre name are on them and they are securely fastened to this booklet.

Please complete the details below clearly and in BLOCK CAPITALS.

Learner name _____

Centre name _____

Learner number _____ Centre number _____

Do not turn over until the invigilator tells you to do so.

SAMPLE

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SAMPLE

Activity 1: Flat share

1 (a) Alex, Jas and Stef are students. They are looking for a flat to share.

They see these adverts online:

3 bedroom flats in Upton:
£433 per month
£450 per month
£395 per month
£475 per month
£389 per month
£375 per month

3 bedroom flats in Downton:
£368 per month
£425 per month
£412 per month
£330 per month
£447 per month

What percentage of the flats in Downton cost over £420 per month?

[1 mark]

Your answer: **%**

1 (b) They want to know which location is the cheapest.

Use the median monthly rental prices to show which location is the cheapest.
Show your working.

[3 marks]

Your answer: **£**

Please turn over

SAMPLE

1 (c) Alex, Jas and Stef each get a student loan to help with living expenses.

They decide to allocate two fifths of their loans for food, and one-sixth for travel.

What fraction of their student loan will be left to spend?

[2 marks]

Your answer:

SAMPLE

1 (d) The students work out that together they have £5400 each year to spend on rent.

They find a flat to rent that will cost £5200 per year.

If the rent **increases by 2%** next year, will it still be within their budget?
Show your working.

[2 marks]

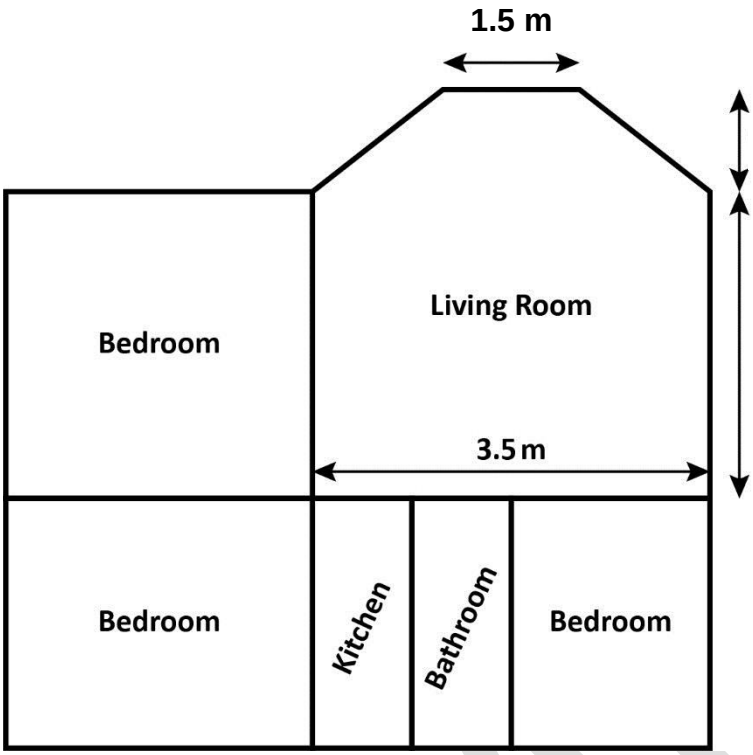
Your answer:

SAMPLE

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1 (e) The plan shows a flat the students like.

The living room has a trapezium-shaped bay window.

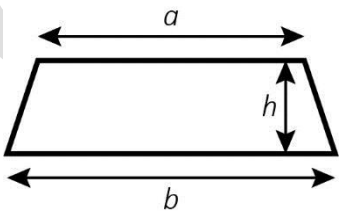


Not drawn accurately

The landlord will replace the carpet in the living room with new flooring, as long as it costs £250 or less.

The new flooring costs £20 per m^2

The formula for finding the area of a trapezium is: $\frac{(a+b)}{2}\ h$



Will the landlord replace the carpet with the new flooring?
Explain your answer.

[4 marks]

SAMPLE

Please turn over

1 (f) Alex, Jas and Stef decide to rent a different 3 bedroomed flat.

They choose their bedrooms.

- Alex's bedroom has an area of 12 m^2
- Jas's bedroom has an area of 18 m^2
- Stef's bedroom has an area of 20 m^2

They decide to share the rent between them:

Option 1 – Share the rent **equally** between them
Option 2 – Share the rent **based on the area** of their bedrooms.

Which option will work out cheapest for Jas?
Show your working.

[3 marks]

Your answer: **Option**

[Total marks: 15]

This is the end of Section A.

SAMPLE